

STRATA: Catastrophic Forgetting as an Architectural Default, a Measured Failure Taxonomy, and a Replay-Free Resolution at Prototype Scale

Christopher Gardner*

July 2026

Abstract

Catastrophic forgetting is conventionally treated as a training pathology to be suppressed with regularizers or replay. We argue, and demonstrate at prototype scale, that it is instead the *default consequence* of three architectural decisions most networks make implicitly: writing new information into the same substrate as old information, at the same timescale, in a correlated basis. We present STRATA (Sparse Tiered Routing with Anchored Two-timescale Adaptation), a continual-learning architecture that makes all three decisions differently, using only local learning rules: no backpropagation through time, no task labels, and no replay buffers of raw data. Seven composable mechanisms—sparse hysteretic routing, atomic feature allocation, a fast/slow synaptic split, multi-timescale cascade consolidation, maturity-gated readouts, predictive-residual coding, and inference-side decorrelation with impermanent consolidation—are each pinned to a specific, empirically measured failure mode. On sequential task benches with forced cross-task interference, STRATA exhibits flat retention over 24–48 sequential tasks, capacity growth that tracks task *diversity* rather than task count (16 columns for 48 tasks across 16 families), up to $2.6\times$ forward transfer, and graceful power-law degradation; with representation learning enabled, the learned dictionary ultimately *outperforms* a fixed random expansion on the primary bench. We further embed the architecture as a metabolically constrained organ in an artificial organism, where learning generates heat that throttles its own plasticity, and demonstrate a two-layer stack in which event-driven ascent and bounded top-down priors reduce regime re-orientation transients by 15–31%; the same experiments yield a measured design constraint for hierarchy—abstraction density at level $N+1$ requires geometrically more experience than level N can supply per unit time. Alongside the positive results we report a taxonomy of failure modes—absorption, code drift, correlation inheritance, capacity exhaustion, orphaned bindings, threshold collapse, and prior capture—each discovered by measurement, several of which recurred at higher levels of the hierarchy in level-specific form, and a falsification record in which every mechanism was tested against a pre-registered failure criterion. We state the scope limits plainly: all results are at toy scale on self-designed benches, and composition under depth beyond two layers, external benchmarks, and order-of-magnitude scaling remain open.

1 Introduction

This paper’s central claim is that “catastrophic forgetting” is not one phenomenon but several: an umbrella term for a set of distinct, individually measurable failure modes—*absorption* of an old

*Correspondence: christhechef35@gmail.com. Architecture design, implementation and experiments were carried out in an interactive research session with Claude (Anthropic); all results reported here are reproducible from the accompanying code.

module’s identity by new data, *code drift* invalidating downstream bindings, *correlation inheritance* in learned features, *capacity exhaustion* of finite dictionaries, and *orphaned episodic bindings* under representation change—each of which can be detected by instrumentation, prevented by a specific mechanism, and unit-tested. We present this as a *candidate* decomposition: it is internally predictive within the systems studied here (§7), and whether it is the structure of the phenomenon at large—or the first chart of it—is precisely what evaluation on benchmarks we did not design must decide. What the decomposition does establish is that, in the systems studied, forgetting is the default consequence of three architectural decisions most networks make implicitly: writing new information (i) into the same physical substrate as old information, (ii) at a single timescale of plasticity, and (iii) in a representational basis whose components are correlated across experiences.

Modern neural networks learn well from stationary, shuffled data and forget catastrophically from sequential, non-stationary data [3, 10]. The dominant remedies—elastic regularization of important weights [6, 15], replay of stored or generated past data [14], or frozen task-specific expansion [13]—treat forgetting as a defect of the *training procedure*. Under the decomposition above they are partial by construction: a loss term attacks only the timescale condition, and replay converts the non-stationary problem back to a stationary one at the cost of storing data, while the interference both penalize is manufactured upstream of the loss.

We test the hypothesis constructively. STRATA is a small, fully-instrumented continual-learning system in which each of the three decisions above is made differently, by mechanisms with independent support in theoretical neuroscience: complementary learning systems [9], cascade models of synaptic consolidation [4, 1], sparse expansion coding [11, 2], predictive coding [12], and reservoir computing [7, 5]. The contribution is not any single mechanism—most have antecedents—but (a) evidence that they *compose* into one working system under strict constraints (local rules only, no replay, no task labels, bounded state), and (b) a *measured failure taxonomy*: a catalogue of the specific ways such systems break, each observed in instrumented experiments, each answered by a named mechanism, and several observed to recur at higher levels of a hierarchy in new costume. We believe the taxonomy is the most transferable artifact, because these failures will confront any architecture attacking this problem at any scale.

Contributions.

1. An architectural account of catastrophic forgetting (substrate, timescale, basis) and a seven-mechanism blueprint that operationalizes it with purely local learning rules (§3).
2. A measured failure taxonomy for replay-free continual learning (§4), including three failure modes we have not seen named elsewhere: *absorption* (identity capture without weight erasure), *correlation inheritance* in learned dictionaries, and *orphaned episodic bindings* under representation drift.
3. Benchmarks with *forced* interference (shared vocabulary, conflicting successors), motivated by the measured observation that disjoint-task benches are jointly fittable and produce no forgetting even in unprotected baselines (§5).
4. Empirical results: flat retention across 24 sequential tasks; capacity growth tracking diversity, not task count; 2.6× forward transfer; and a learned dictionary that ultimately beats a fixed random expansion under continual-learning constraints (§6).
5. A metabolic embedding in which synaptic work is paid for in a conserved energy economy, producing an emergent learn/overheat/rest cadence and reproducing a known saturation pathology of the host system (§6.5).

6. A two-layer stack with event-driven ascent and bounded top-down priors, reducing regime re-orientation transients by 15–31%, together with negative results delineating when hierarchy does *not* pay (§6.7).

2 Forgetting as an Architectural Default

Consider online learning of a predictive map from a non-stationary stream. Interference between a new experience and consolidated knowledge requires three simultaneous conditions: the update must touch *the same parameters* (substrate overlap), it must move them *faster than they can be protected* (timescale collision), and the representations involved must *overlap* (basis correlation). Standard architectures satisfy all three by default: dense representations make every parameter serve every input; a single learning rate makes protection and adaptation the same knob; and learned features inherit whatever correlations exist in the input distribution. Regularization methods attack the middle condition only, and replay methods convert the non-stationary problem back to a stationary one at the cost of storing data. STRATA instead removes each condition architecturally: routing and sparsity decide *where* information is written; a fast/slow split and consolidation cascades decide *when*; and predictive-residual coding decides *what*—only the deviation from a slow expectation ever reaches plastic weights.

3 The STRATA Architecture

STRATA is an online next-input predictor. Inputs $x_t \in \mathbb{R}^d$ arrive one per tick with no task labels and no episode boundaries. All learning rules are local (delta rules and imprinting); the only temporal credit assignment is through bounded liquid state, never through backpropagation through time.

3.1 Routing and the slow pathway

Each *column* c owns a centroid $\mu_c \in \mathbb{R}^d$: an exponential moving average of the inputs it has claimed, serving simultaneously as (i) its routing identity and (ii) its slow *expectation* of the input. A slow context $\kappa_t = (1 - \rho) \kappa_{t-1} + \rho x_t$ is routed by cosine affinity $a_c = \cos(\kappa_t, \mu_c)$ with **three-way hysteresis**:

$a_c \geq \theta_{\text{claim}}$: column may absorb the input into its identity (and update claim statistics)

$a_c \geq \theta_{\text{route}}$: column participates in prediction *and* learning

$a_c < \theta_{\text{route}}$: column may serve as fallback predictor but receives *zero* learning,

with $\theta_{\text{claim}} \gg \theta_{\text{route}}$ and all identity updates frozen while a regime transition is detected. Persistent orphanhood (no column routable), measured by a leaky accumulator rather than a consecutive-tick counter, spawns a new column warm-started from its nearest neighbour. Optionally, thresholds are *adaptive*: per-column claim statistics (m_c, v_c) tracked only on claimed ticks give $\theta_{\text{claim},c} = \max(m_c - \lambda\sqrt{v_c}, b + \kappa\sigma_b)$, where (b, σ_b) are background-similarity statistics tracked on orphaned ticks; the floor term is load-bearing (§4, *threshold collapse*).

3.2 Predictive-residual coding (the two-pathway split)

The plastic pathway never sees the raw input. With μ the claiming column’s centroid, the residual $r_t = x_t - \mu$ is encoded by a sparse expansion $z_t = \text{enc}(r_t) \in \mathbb{R}^D$ (s -sparse, $s \ll D$), and the final

prediction adds the expectation back:

$$\hat{x}_{t+1} = \mu + \underbrace{\sum_c g_c (C_c h_t^c + D_c z_t + Q_c q_t)}_{\text{slow readouts}} \oplus \underbrace{F z_t}_{\text{fast weights}}, \quad (1)$$

where $\oplus$ denotes confidence-weighted blending. Because the shared (correlated) component of a regime is absorbed by μ , the dictionary codes only deviations—removing, by construction, the input correlations that learned features were measured to inherit fatally (§6.3).

3.3 Liquid state and readouts

Each column carries a fixed, diagonal, input-dependent leaky reservoir with *liquid* time constants stretched by novelty ν_t :

$$h_t^c = e^{-1/\tau_c(x_t, \nu_t)} \odot h_{t-1}^c + (1 - e^{-1/\tau_c(x_t, \nu_t)}) \odot (W_B x_t), \quad \tau_c = \tau_{\min} + \text{softplus}(w_\tau^\top x_t) (1 + \alpha \nu_t). \quad (2)$$

The decay factor lies in $(0, 1)$ elementwise, so the state is bounded for all time. Three linear readouts learn by local delta rules: C_c over the state, D_c over the code, and Q_c over a fixed random k WTA *conjunctive expansion* $q_t = \text{kWTA}_S(R[z_t \oplus h_t^c])$, which manufactures symbol-in-temporal-context features and lifts effective key rank above low-rank input manifolds at zero churn risk (nothing in R ever moves).

3.4 The fast/slow split and consolidation

A fast-weight matrix F binds keys to residual targets by a novelty-gated delta rule, $F \leftarrow F + \eta_t^w (v_t - F z_t) z_t^\top$, capturing surprising experience one-shot with no gradient. During calm periods, anchors sampled from F 's own content are distilled into the claiming column's slow readout: consolidation happens later, from the system's own memory, never from stored raw data. Every slow parameter is the surface of a Benna–Fusi diffusion chain $u_1, \dots, u_m$ with geometrically spaced timescales; learning writes only u_1 , and symmetric inter-level exchange converts exponential forgetting into power-law retention with relearning savings [1, 4].

3.5 Learning the dictionary

The encoder's rows may themselves learn, under three protections: **(i) per-unit consolidation**, $\eta_i = \eta_0 / (1 + \kappa \text{usage}_i)$; **(ii) novelty recruitment with atomic allocation**: when no row matches a residual, the least-used row is drafted and *imprinted* one-shot on the Gram–Schmidt residual of the input against co-active mature rows, and the input is *re-encoded before anything downstream binds to it*, so keys are born stable; and **(iii) usage decay**, which makes consolidation impermanent so a finite dictionary forgets gracefully, least-used first. Inference may run either as k WTA or as *matching pursuit* [8]: each selected atom's contribution is subtracted before the next selection, which performs lateral decorrelation *by inference* rather than by weight dynamics and therefore cannot conflict with consolidation. Slow readouts consolidate onto a key only as fast as the key's supporting rows stop moving (*maturity gating*). Dictionaries may also grow structurally: capacity added under allocation pressure is *dormant*—masked out of inference entirely until the draft-and-imprint path activates it—which makes growth exactly backward-transparent (bit-identical old codes, unit-tested).

3.6 The master signal

A single standardized surprisal $\nu_t = \|x_t - \hat{x}_t\|^2 / \widehat{\text{Var}}$ gates everything: fast-weight writes open with novelty; slow learning closes with it ($1/(1 + \max(0, \nu_t - 1))$), so surprising samples are captured episodically rather than consolidated; a CUSUM statistic on ν_t declares regime transitions, freezing identity updates and damping mature columns; and sustained routing orphanhood drives structural growth.

4 A Measured Failure Taxonomy

Every entry below was observed in an instrumented experiment before being named; each is prevented by a specific mechanism and, where marked ($\uparrow$), was observed to *recur at the next level of the hierarchy* in level-specific form.

Failure	Signature (measured)	Countermeasure
Absorption $\uparrow$	A new task drags an old column’s centroid onto itself; affinity never dips below threshold (0.62 minimum at a boundary; 0.92 across a schedule change at layer 2)	Claim $\gg$ route hysteresis; identity frozen during transitions; fallback predicts but never learns
Code drift	Learned encoder rotation invalidates downstream keys; old-task error $0.23 \rightarrow 0.76$	Per-unit consolidation; novelty recruitment; two-pathway residual coding
Correlation inheritance	Learned features fire for every correlated sibling; within-task code separation $ \cos 0.21 \rightarrow 0.33\text{--}0.45$	Residual coding (shared structure never reaches the dictionary); matching-pursuit inference
Orphaned bindings $\uparrow$	Episodic keys formed pre-allocation never match post-imprint codes	Atomic allocation: re-encode after imprint; layer boundaries transmit only post-settlement codes
Capacity exhaustion	All virgin rows consumed (288 symbols vs. 128 units); floors degrade $0.168 \rightarrow 0.258$ while a random encoder stays flat	Usage decay (graceful, least-used-first); dormant structural growth
Threshold collapse	A column claiming noise widens its own adaptive band; absorption returns through the statistics	Background floor $\theta_{\text{claim}} \geq b + \kappa\sigma_b$; statistics update only on claimed ticks
Prior capture	A confident-but-wrong top-down prior forces the lower layer into the wrong regime	Bounded prior bonus (can promote a routable column, never make an alien one routable); priors bias inference, never learning
Consolidation misattribution	Key-affinity anchor assignment distilled one task’s memories into another task’s column	Ownership from the routing context, not key similarity

Table 1: The failure taxonomy. Numbers reference the experiments of §6.

5 Benchmarks

All benches stream noisy cyclic sequences of unit vectors ($d=24$, cycle lengths 12–13, additive noise $\sigma=0.02$), one sample per tick, with tasks switched without any signal to the learner. A methodological finding shaped their design: **disjoint-vocabulary task sequences produce no measurable forgetting even in unprotected baselines**, because a single linear map can fit all tasks jointly whenever no key maps to two targets and capacity suffices. Forgetting must be *forced*: our tasks share a vocabulary subset whose successors differ per task (interference bench), or belong to *families* with correlated symbols and partially shared transitions (long-sequence bench; 8 families $\times$ 3 variants = 24 tasks), optionally with family residuals confined to a rank-3 subspace (low-rank bench). Retention is measured by *frozen-circuit probes*: a deep copy of the system with all plasticity, fast-weight capture, and identity updates disabled, evaluated on a past task after the full schedule. Forward transfer is measured as time-to-criterion on a family’s later variants relative to its first.

6 Results

6.1 Layered defenses on the interference bench

Stripping mechanisms one at a time (600 ticks per phase, schedule $A \rightarrow B \rightarrow C \rightarrow A$) yields post-transient recall error on task A of **0.18** (full), 0.34 (no routing), 0.32 (no routing or cascade), and **0.82** (dense unprotected baseline); relearned error returns to the originally-learned level (0.042 vs. 0.045) for the full system. A frozen-circuit probe of an earlier configuration showed task A’s circuit degrading only $0.085 \rightarrow 0.132$ across 1200 ticks of other tasks: graceful power-law sag, not erasure.

6.2 Long-sequence bench: retention, capacity, transfer

Table 2 shows the full-rank bench (24 tasks, 500 ticks/task). Capacity grew to 8 columns for 24 tasks across all conditions—tracking the 8 families, i.e. *diversity*, not task count. Retention is flat with age (oldest vs. newest third). Later family variants reach criterion up to $2.6\times$ faster than first variants.

Encoder	TTC v1	TTC v2+	Fwd	Ret. (old)	Ret. (new)	Cols
Fixed random	197	87	$2.3\times$	0.214	0.216	8
Fixed + MP	298	159	$1.9\times$	0.253	0.255	8
Learnable, naive	240	129	$1.9\times$	0.694	0.434	8
Learnable, consolidated	220	96	$2.3\times$	0.217	0.224	8
Consolidated + MP	232	88	$2.6\times$	0.216	0.209	8
Consolidated + MP + decay	220	89	$2.5\times$	0.229	0.219	8

Table 2: Full-rank long-sequence bench (24 tasks; TTC = ticks to criterion; Ret. = frozen-circuit error, oldest/newest third of tasks). The learned dictionary with matching pursuit is the best overall system; the fixed + MP control confirms the mechanism (explaining away pays only when atoms are learned to be worth explaining with).

6.3 The representation-learning arc

Enabling encoder learning naively reintroduced catastrophic aging (old-task retention error 0.69 vs. 0.21 fixed). Per-unit consolidation with novelty recruitment rescued retention (0.29) but *con-*

sistently destroyed forward transfer ($0.9\times$) across three allocation variants (incremental, one-shot imprinting, residual imprinting). Instrumentation localized the cause: learned features *inherit input correlations* that random projections destroy—within-task code separation degraded from $|\cos| = 0.21$ (random) to $0.33\text{--}0.45$ (learned)—and the textbook remedy, continuous anti-Hebbian decorrelation, would perpetually move mature features, i.e. exactly the churn consolidation exists to prevent. **The stability–plasticity dilemma re-emerges one level down, at the representation layer, and consolidation alone does not solve it there.**

What solved it was architectural: the two-pathway residual code. With the shared component absorbed by column expectations, the dictionary sees only near-orthogonal residuals; the same consolidated learner then reached parity with the fixed encoder on all task metrics with *better* code geometry (0.036 vs. 0.054), and, after acquisition was decoupled from allocation (atomic re-encoding plus maturity gating), surpassed it (Table 2). The whole system also improved: acquisition time halved ($427 \rightarrow 197$) and transfer rose from $1.6\times$ to $2.3\times$ even for the fixed encoder.

6.4 Low-rank structure, capacity limits, and expressivity

Confining family residuals to a rank-3 subspace makes symbol directions inherently correlated. Matching-pursuit inference plus the learned dictionary beat the random encoder where theory requires—single-task error floor 0.168 vs. 0.188 ; code separation 0.048 vs. 0.137 —until **capacity exhaustion**: with 288 symbols against 128 units, all virgin rows were consumed within two of eight families, after which floors degraded ($0.168 \rightarrow 0.258$) while the random encoder stayed flat (nothing to exhaust; mediocre everywhere, forever). Usage decay converted the cliff into a graceful trade-off (transfer $0.9\times \rightarrow 1.2\times$ at fixed capacity). The residual transfer gap versus the fixed encoder was then closed by the conjunctive expansion: on rank-3, learned+MP improved from $1.2\times$ to $1.7\times$ forward transfer (TTC on later variants $381 \rightarrow 155$), matching the fixed encoder—confirming the gap had been readout expressivity, not representation quality.

6.5 Metabolic embedding

STRATA updates are outer products, so the Frobenius norm of every weight change is available at no cost; the host organism charges this synaptic work against a conserved energy economy in which work generates *load* (heat), load raises entropy, entropy degrades fidelity, and fidelity gates the next tick’s plasticity. Two results: (i) at a standard energy capacity, baseline cognitive work alone pinned load above the learning-reachability ceiling—the organ entered a saturated no-learning attractor (accuracy 0.12 ; learning unreachable on $397/400$ ticks), reproducing a known pathology of the host system in a learning organ; (ii) with a brain-sized energy budget, a **burst/cool cadence emerged from pure physics**: learning-reachability flickered at ~ 4 -tick spacing (86 ceiling crossings in 400 ticks) with no cooling rule anywhere in the code, at an accuracy cost relative to the unconstrained control (0.34 vs. 0.57) attributable to a separately documented saturated-fidelity attractor of the host. Methodologically, the energy instrumentation acted as an autonomic bug detector: a silent learning collapse (a missing normalization) was caught because the organ went *cold*.

6.6 Scale verification at 48 tasks

Doubling the schedule (16 families $\times$ 3 variants, rank-3 residuals, 400 ticks/task) with the growing dictionary (initial size 64) and adaptive claim bands produced the falsification cycle recorded in Table 3: the first run spawn-stormed (48 columns for 48 tasks; newest-third frozen-probe error above 1.0 because tasks arriving after the column cap were orphaned into no-learning fallback), the cause was traced to background statistics absorbing transition swings, and after the statistics-freeze

fix both a static-threshold control and the adaptive retest produced **exactly 16 columns for 16 families in every condition**—diversity scaling intact at twice the task count—with retention flat with age (best row: 0.306 oldest vs. 0.310 newest third) and forward transfer of 1.4–1.6 $\times$. Adaptive bands slightly outperformed static thresholds after the fix (e.g., fixed encoder: 0.306 vs. 0.334 old-task retention). One honest finding, settled by a pre-committed reading rule (any ambiguous outcome at under-demand capacity mandates a rerun clearing demand before conclusions): at dictionary capacities of 128, 512, and 768 units—the last comfortably clearing the 576-symbol allocation demand—the fixed encoder remains the best overall row on rank-3 at 48 tasks. The gap is therefore **capacity-independent**: the learned-dictionary advantage of Table 2 is established on isotropic residuals and does not extend to low-rank residual structure at this scale. Localizing why is open work.

6.7 Depth: a two-layer stack

Layer 2 ticks only at Layer 1’s regime boundaries: each detected transition, after settlement, is pooled into one event $u = \sum_t \nu_t \phi(z_t, e_t) / \sum_t \nu_t$, and L2 runs the identical mechanism suite over events. Its predictions decode into bounded routing priors and context seeding at the next transition. Across three seeds the revisit transient (error in the first 40 ticks of a returning regime) dropped **15–31%** (mean $\approx 23\%$). Three lessons: (i) *emit what arrived, not what confusion felt like*—pooling the surprise burst itself made all boundaries look identical and forbade L2 differentiation (fixing this raised the payoff from 18% to 31%); (ii) absorption recurred at L2 as context-centroid co-drift over a shared event stream (affinity 0.92 across a complete schedule change), with the additional geometric fact that *mixtures of unit vectors crowd together*, making mixture-level identity intrinsically thin-margined; and (iii) a measured **design constraint for hierarchy** that we propose as a general law: *hierarchical abstraction density requires geometric timescale separation*. L2 can differentiate regime-arrival types (2–3 columns tracking schedule structure under event-granularity thresholds), but at tens of events per run, splitting experience across columns starves each successor map and prior quality degrades (31% $\rightarrow$ 10%). The system thereby identifies, correctly, that the stream does not yet contain enough regime-level experience to support differentiated regime-level structure—the best overseer at small event budgets is the least differentiated one—and quantifies how much more it would need. Abstraction at level $N+1$ requires orders of magnitude more time than level N ; schedule-level structure belongs to a third layer, not to a threshold adjustment at the second.

7 The Falsification Record

Each mechanism was tested against a failure criterion stated *before* the experiment ran. We report the record in full, including the entries where the design as first built failed its own test and had to be recalibrated—these entries are, in our view, the strongest evidence that the taxonomy is doing real work rather than post-hoc justification.

8 Related Work

Regularization approaches estimate parameter importance and slow important weights [6, 15]; STRATA’s cascade and per-unit consolidation play a similar role but are computed online from local signals, with no Fisher estimation and no task boundaries, and are demonstrated to be insufficient alone at the representation layer (§6.3). Replay methods [14] revisit stored or generated data;

STRATA’s calm-time distillation replays only its own fast-weight content. Structural methods [13] isolate tasks in grown modules; STRATA’s columns are similar in spirit but are recruited by measured orphanhood rather than task labels, grow with diversity rather than task count, and share a learned encoder. The fast/slow organization follows complementary learning systems [9]; the cascade follows [4, 1]; sparse expansion coding follows [11, 2]; residual coding follows predictive coding [12]; matching pursuit is classical [8]; fixed high-dimensional dynamics with learned readouts follow reservoir computing [7, 5]. Our contribution is the composition under strict continual-learning constraints, and the failure taxonomy that the composition exposed.

9 Limitations

We state these plainly. (1) **Scale**: all results are at $d=24$, ≤ 48 tasks, ~ 300 symbols, linear readouts over fixed expansions; per-tick dynamics are interpreted, not optimized. (2) **Benchmark ownership**: every bench was designed by us; there is no comparison against EWC, replay, or progressive networks on benchmarks we do not control. A pre-registered protocol for that comparison—including the interventional criterion for what observed forgetting would count as *outside* the trichotomy, fixed before any external run—accompanies the code ([paper/preregistration.md](#)). (3) **Depth**: one representation layer plus a two-layer routing hierarchy; whether the stability machinery composes through deep feature hierarchies is untested and is, in our judgment, the single most important open question. (4) **Hand-set constants**: routing thresholds were tuned to measured geometry; the adaptive-band mechanism *failed its first pre-registered scale falsifier* (spawn storm; Table 3) and passed revalidation only after the statistics-freeze redesign (§6.6); further scale remains untested. (5) **Capability ceiling**: stability was purchased with deliberately weak learners; the learned-versus-fixed dictionary advantage, while now positive, is modest. (6) Some forgetting is information-theoretically mandatory at finite capacity on unbounded streams; STRATA chooses *which* (graceful, least-used-first), not *whether*.

10 Conclusion

Treated as an architectural default rather than a training pathology, catastrophic forgetting decomposes into engineering constraints that can be named, measured, unit-tested, and removed. Seven local mechanisms suffice, at prototype scale, for label-free, replay-free continual learning with flat retention, diversity-scaled capacity, positive forward transfer, graceful forgetting, metabolic self-regulation, and—after resolving the representation-layer dilemma architecturally—representation learning that helps rather than harms. The failure taxonomy generalized upward through one level of hierarchy, in level-specific costume, exactly as a useful theory should: it told us where to look when things broke. The decisive next experiments are composition under depth, benchmarks we do not own, and an order of magnitude of scale.

Reproducibility. All experiments are reproducible from the accompanying code: `python -m strata.demo` (interference bench), `python -m strata.longbench` (long-sequence bench; `--rank 3`, `--conjunctive 512`, `--grow-from`, `--adaptive`), `python -m strata.stack` (depth bench), `python main.py --cortex [--thermo]` (metabolic embedding), and `pytest` (62 tests, including exact backward-transparency of structural growth and the zero-plasticity freeze contract).

References

- [1] M. K. Benna and S. Fusi. Computational principles of synaptic memory consolidation. *Nature Neuroscience*, 19:1697–1706, 2016.
- [2] S. Dasgupta, C. F. Stevens, and S. Navlakha. A neural algorithm for a fundamental computing problem. *Science*, 358(6364):793–796, 2017.
- [3] R. M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4):128–135, 1999.
- [4] S. Fusi, P. J. Drew, and L. F. Abbott. Cascade models of synaptically stored memories. *Neuron*, 45(4):599–611, 2005.
- [5] H. Jaeger. The “echo state” approach to analysing and training recurrent neural networks. GMD Technical Report 148, 2001.
- [6] J. Kirkpatrick et al. Overcoming catastrophic forgetting in neural networks. *PNAS*, 114(13):3521–3526, 2017.
- [7] W. Maass, T. Natschläger, and H. Markram. Real-time computing without stable states. *Neural Computation*, 14(11):2531–2560, 2002.
- [8] S. Mallat and Z. Zhang. Matching pursuits with time-frequency dictionaries. *IEEE Transactions on Signal Processing*, 41(12):3397–3415, 1993.
- [9] J. L. McClelland, B. L. McNaughton, and R. C. O’Reilly. Why there are complementary learning systems in the hippocampus and neocortex. *Psychological Review*, 102(3):419–457, 1995.
- [10] M. McCloskey and N. J. Cohen. Catastrophic interference in connectionist networks. *Psychology of Learning and Motivation*, 24:109–165, 1989.
- [11] B. A. Olshausen and D. J. Field. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*, 381:607–609, 1996.
- [12] R. P. N. Rao and D. H. Ballard. Predictive coding in the visual cortex. *Nature Neuroscience*, 2(1):79–87, 1999.
- [13] A. A. Rusu et al. Progressive neural networks. *arXiv:1606.04671*, 2016.
- [14] H. Shin, J. K. Lee, J. Kim, and J. Kim. Continual learning with deep generative replay. *NeurIPS*, 2017.
- [15] F. Zenke, B. Poole, and S. Ganguli. Continual learning through synaptic intelligence. *ICML*, 2017.

A Hyperparameters

Default configuration (dimensionless units; one sample per tick): input $d=24$; code $D=128$, sparsity $s=8$; state $n=32$ per column; $\rho_{\text{context}}=0.08$; $\theta_{\text{route}}=0.65$, $\theta_{\text{claim}}=0.8$; spawn: leaky rate 0.15, threshold 0.8; cascade: 4 levels, $\tau_1=8$, ratio 4; learning rates: state readout 0.05, code readout 0.3,

distillation 0.1, fast weights 0.6 (novelty-gated, hard floor at 5% of maximum rate); CUSUM slack 0.5, threshold 8, transition refractory 15 ticks, mature-column damping 0.15; encoder learning 0.15 with consolidation $\kappa=0.2$, recruitment threshold 0.5, usage decay 2×10^{-4} (when enabled); conjunctive expansion 512 with $S=12$ (when enabled); layer-2: event-driven, settle window 15 ticks, $\rho=0.12$, $\theta_{\text{claim}}=0.9$, cascade τ_1 offset $\times 16$.

B On the use of failure-driven development

Ten of the mechanisms reported here were not in the original design; each was added after an instrumented experiment falsified the design as it stood, and three (absorption hysteresis, dormant growth, post-settlement event pooling) were forced by failures that the a-priori mathematics gave no reason to expect. We record this because it bears on methodology: in this domain, contact with data was worth more than derivation at every single decision point.

Mechanism	Pre-registered hypothesis	Falsifier	Outcome
Naive encoder learning	Unprotected representation learning will reintroduce forgetting (stated before implementation)	Retention remains flat	<i>Hypothesis confirmed:</i> old-task error $0.23 \rightarrow 0.76$
Consolidation alone	Per-unit consolidation suffices for a learnable encoder	Transfer or retention degrades	Falsified three times (three allocation variants): retention rescued, transfer destroyed ($0.9\times$); cause measured as correlation inheritance
Two-pathway residual coding	Removing shared structure before the dictionary resolves correlation inheritance	Code separation fails to recover	Survived: $ \cos $ $0.33\text{--}0.45 \rightarrow 0.036$, better than random (0.054); all task metrics improved
Conjunctive expansion	The rank-3 transfer gap is readout expressivity, not representation quality	Gap fails to close with superior codes	Survived decisively: $1.2\times \rightarrow 1.7\times$, TTC $381 \rightarrow 155$
Structural growth	Zero-initialized growth is backward-transparent	Any existing code or readout changes	Falsified as designed (fresh random rows won inference for old inputs); redesigned with dormant capacity; then survived bit-exactly (unit-tested)
Two-layer stack	Top-down priors cut the revisit transient	No measurable reduction	Survived (15–31%), after one interface recalibration: pooling the surprise burst fed L2 indiscriminative events (18%); pooling post-settlement arrivals fixed it (31%)
L2 diversity scaling	L2 columns track schedules as L1 columns track tasks	Column count fails to track schedule diversity	Falsified at current event budgets ; refined into the geometric-timescale design constraint (§6.7)
Adaptive bands at scale	Statistics-tracked thresholds preserve diversity scaling at 48 tasks / 16 families	Spawn storm or threshold collapse	Falsified as designed : spawn storm (48 columns for 48 tasks; family sharing collapsed; late-task retention pathological). Cause traced to background statistics updating during transition swings, inflating the adaptive floor. Fix: statistics freeze during transitions (the same law as centroids and claim statistics). <i>Revalidation passed</i> : 16 columns for 16 families, matching a